A Text Has No Taste: Synesthetic Probing of Language Models and the Absent Referent

Omar Cherif
LORIA
mohamed.cherif@inria.fr

Christophe Cerisara
LORIA
christophe.cerisara@loria.fr

Julien Falgas
CREM, Université de Lorraine
julien.falgas@univ-lorraine.fr

Abstract

We ask a language model whether a text is *purple*, *salty*, *grainy*, *nasal* or *smoky*, and read the difference between the logits of "yes" and "no" as a feature. With only twenty such *synesthetic* questions spanning the five senses, a linear classifier reaches 0.93 accuracy on sentiment (IMDB) and beats a random-word control by large margins on four datasets. Crucially, we manipulate the *metaphoricity* of the questions: predicates such as *purple* or *umami* carry no conventional figurative meaning, unlike *dark* or *bitter*. While more metaphorical predicates are individually somewhat more useful, low-metaphoricity sensory probes—which can only be answered by reference to a perception the model never had—remain strongly informative on every task. We read this through Saussure: a text-only model has no *referent*, so sensory predicates are unconstrained signifiers whose value comes from differences within the language system, not from the senses. A red text and an acidic text are, for the model, equally sayable.

1 Introduction

A language model can be asked whether a text is "acidic", "rough" or "purple". There is no fact of the matter: a string of characters has no taste, no texture and no colour. Yet the model never refuses to answer, and—as we show—its graded answers carry reliable information about the text. This paper takes that small observation seriously.

We extend the feature-generation approach of Florescu and England, in which an LLM is queried with simple questions about a text and the answer logits are used as features for a downstream classifier. Out of the many question types one could ask, we isolate the most theoretically puzzling one: *synesthetic* questions, which describe a text through a sensory modality (Cytowic). Because the model perceives only text, such questions cannot be grounded in perception; they should be meaningless. They are not.

Our contributions are: (i) a controlled set of 20 synesthetic questions (four per sense) selected for *low metaphoricity*, i.e. minimal conventional figurative meaning, together with a matched high-metaphoricity set; (ii) an evaluation on four datasets spanning sentiment, topic and emotion, showing that synesthetic features classify far above a random-word control and chance, and rival TF–IDF on affective tasks; (iii) an analysis showing that usefulness only weakly depends on metaphoricity—non-figurative sensory probes still work—and (iv) a reading of these results through Saussure's notion of the sign without a referent.

2 Background and related work

Synesthesia in perception and in language. Synesthesia is a phenomenon in which a stimulus in one sensory modality triggers a perception in another (Cytowic). Language is itself full of *lexicalized* synesthetic transfers—a *sweet* voice, a *warm* colour—which follow regular cross-modal directionalities (Ullmann, 1957; Strik Lievers, 2015; Winter, 2019). These conventional metaphors are precisely what we want to control for: a feature that works because "bitter" already means "resentful" tells us about lexicalized semantics, not about synesthetic projection.

Meaning, form and grounding. Whether a system trained only on form can acquire meaning is contested (Bender and Koller, 2020; Bender et al., 2021; Bisk et al., 2020), and connects to the older symbol grounding problem (Harnad, 1990). Our probe offers an empirical angle: sensory predicates are the clearest case of words whose meaning seems to require perceptual grounding, so they are an ideal stress test for "form without referent".

The Saussurean sign. For de Saussure (1916), the linguistic sign unites a signifier and a signified through an arbitrary relation, and the *value* of a sign arises from its differences with other signs in the system (*la langue*), not from any link to an external thing—the referent is bracketed. A distributional model in the sense of Firth (1957) is, in this light, a system of pure differential values. We argue this is exactly the regime in which our synesthetic questions operate.

3 Method

3.1 Synesthetic questions and metaphoricity

We write 20 yes/no questions, four for each of the five senses (sight, taste, touch, hearing, smell). The questions are chosen to have *low metaphoricity*: the predicate has little or no conventional non-physical meaning in English (e.g. *purple*, *umami*, *starchy*, *grainy*, *nasal*, *metallic*). We deliberately avoid predicates that are dead metaphors (*dark*, *bitter*, *warm*, *loud*, *fishy*), and place those instead in a matched *high-metaphoricity* set of 20 questions. A third set of 10 *gibberish* probes ("does this text evoke the words: lantern, gravel, ...") acts as a control for whether *any* probe carries information. All questions are listed in Appendix A.

To measure metaphoricity independently of our own judgments, we also ask the model itself, for each predicate, whether it "is often used figuratively or metaphorically to describe something non-physical", and record the yes/no logit difference. This model-based score cleanly separates the two sets (high-metaphoricity predicates: *warm* $+11.7$, *dark* $+11.4$, *sweet* $+10.9$; low: *starchy* -7.3, *minty* -6.2, *nasal* -6.0), and is used in the analysis of Section 4.

3.2 Feature extraction

Following Florescu and England, we present the text followed by a question and ask for a yes/no answer. Instead of decoding, we read the model's next-token logits at the answer position and define the feature as

$$f_q(t) = \log \sum_{v \in \text{YES}} e^{z_v} \; - \; \log \sum_{v \in \text{NO}} e^{z_v},$$

where z are the logits and YES/NO are the case and whitespace variants of "yes"/"no". Each text is thus mapped to a 20-dimensional synesthetic feature vector. We use `google/gemma-4-E4B-it` (Gemma Team, Google DeepMind, 2026), a recent instruction-tuned model, with greedy single-pass forward computations (no sampling).

	chance	LOW	HIGH	ALL	GIB
IMDB	.500	.880	**.934**	.924	.660
AG News	.250	.658	.662	**.710**	.558
BBC	.200	.714	.654	**.768**	.504
Emotion	.167	.287	.390	**.424**	.235

Table 1: 5-fold CV accuracy. LOW/HIGH: 20 low-/high-metaphoricity synesthetic questions; ALL: the 40 together; GIB: 10 gibberish probes. TF–IDF for reference: .79/.75/.97/.43.

3.3 Datasets and classifier

We use balanced subsets of 500 examples (stratified by class) from four datasets covering distinct tasks: **IMDB** (sentiment, 2 classes), **AG News** (topic, 4), **BBC News** (topic, 5) and **Emotion** (affect, 6). Texts are truncated to 300 words. On the resulting features we train a logistic regression with standardized inputs and report 5-fold stratified cross-validation accuracy; we deliberately use a weak linear classifier so that performance reflects the information in the features rather than classifier capacity, and we verify that a small MLP gives equivalent results (Table 2). Baselines are the gibberish features, a TF–IDF (1–2-gram) logistic regression on the raw text, direct zero-shot classification, and the majority-class chance level.

4 Results

Synesthetic features classify, gibberish does not. Table 1 and Figure 1 show that the 20 low-polysemy synesthetic features are strongly predictive on every dataset, far above both chance and the random-word control. On IMDB they reach 0.88 (vs. 0.66 for gibberish and 0.79 for TF–IDF); the full 40-question set reaches 0.92. On topic tasks TF–IDF wins (it is, after all, a lexical task), but synesthetic features still recover most of the signal from senses that have nothing to do with topics.

Every sense carries information. Figure 2 breaks the low-polysemy set down by sense (4 features each). On every dataset, every sense is informative above chance—including "smell" and "hearing" for topic classification. Sight is the most informative modality, but no modality is inert. Whatever the model is doing, it is not confined to one privileged sensory axis.

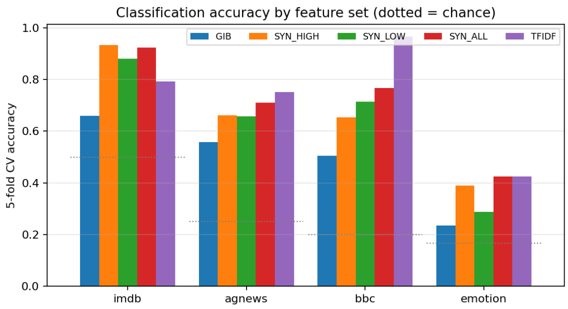

Figure 1: Accuracy by feature set. Synesthetic sets (LOW, HIGH, ALL) clearly exceed the gibberish control (GIB) and chance (dotted) on all four datasets.

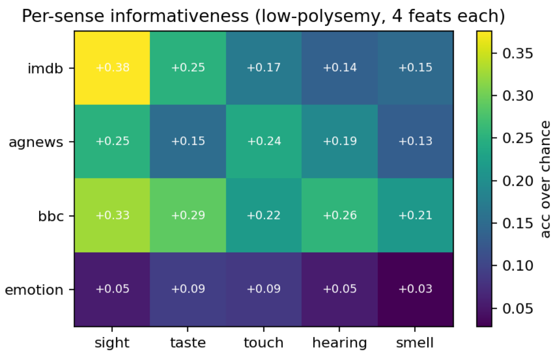

Figure 2: Per-sense informativeness of the low-polysemy probes (accuracy over chance, 4 features per sense). All senses help on all tasks.

		zero	LOW (20)		ALL (40)	
	chance	shot	lr	mlp	lr	mlp
IMDB	.50	.95	.88	.87	.92	.93
AG News	.25	.84	.66	.66	.71	.70
BBC	.20	.92	.71	.71	.77	.78
Emotion	.17	.50	.29	.30	.42	.37

Table 2: Zero-shot direct classification (the model is shown the task) vs. classifiers trained on synesthetic features (which never mention the task), with logistic regression (LR) and a 64-unit MLP. Features recover much of the zero-shot accuracy, and LR≈MLP.

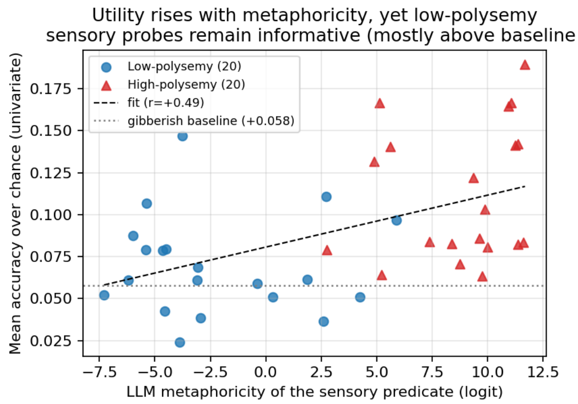

Figure 3: Feature usefulness vs. model-rated metaphoricity. The relation is positive but moderate; low-polysemy probes (blue) stay informative even with no figurative meaning.

Usefulness depends only weakly on metaphoricity. Figure 3 relates each synesthetic question's univariate usefulness (mean over-chance accuracy across datasets) to its model-rated metaphoricity. There is a positive but moderate correlation (Pearson $r = 0.49$, $p < 0.01$; $r = 0.56$ for our a-priori scores): conventionalized metaphors are individually somewhat stronger features, mostly because affective tasks (IMDB, Emotion) reward predicates like *warm* or *bitter*. But the low-metaphoricity predicates—which carry no figurative meaning and could only be answered by an act of perception the model never performed—remain informative, with most lying above the gibberish baseline and the low-polysemy *set* beating gibberish on every dataset (mean over-chance 0.070 vs. 0.058 for single probes, and +0.15 to +0.21 at the set level).

Features are complementary. To check that the synesthetic questions are not merely redundant copies of one another, we add the 20 low-polysemy features in a *random* order and average over 10 orderings (Figure 4). Accuracy rises steadily all the way to the full set on every dataset, and almost every single-feature addition yields an improvement (100% of steps on AG News, 95% on BBC, 89% on IMDB, 79% on Emotion). Each low-polysemy probe thus tends to add information not already captured by the others, rather than re-encoding a single underlying axis.

Comparison with zero-shot. Table 2 compares the trained synesthetic classifiers with direct *zero-shot* classification, in which the model is shown the task and the candidate labels (a closed-set multiple-choice probe). Zero-shot is the natural upper reference and is strong on sentiment and topic. Yet features built only from sensory questions that *never mention the task* recover most of this accuracy—matching zero-shot to within two points on IMDB (.92 vs. .95)—which suggests the synesthetic probes tap the same information the model uses when classifying directly. Logistic regression and a 64-unit MLP give nearly identical numbers, confirming that the signal lies in the features, not in classifier capacity.

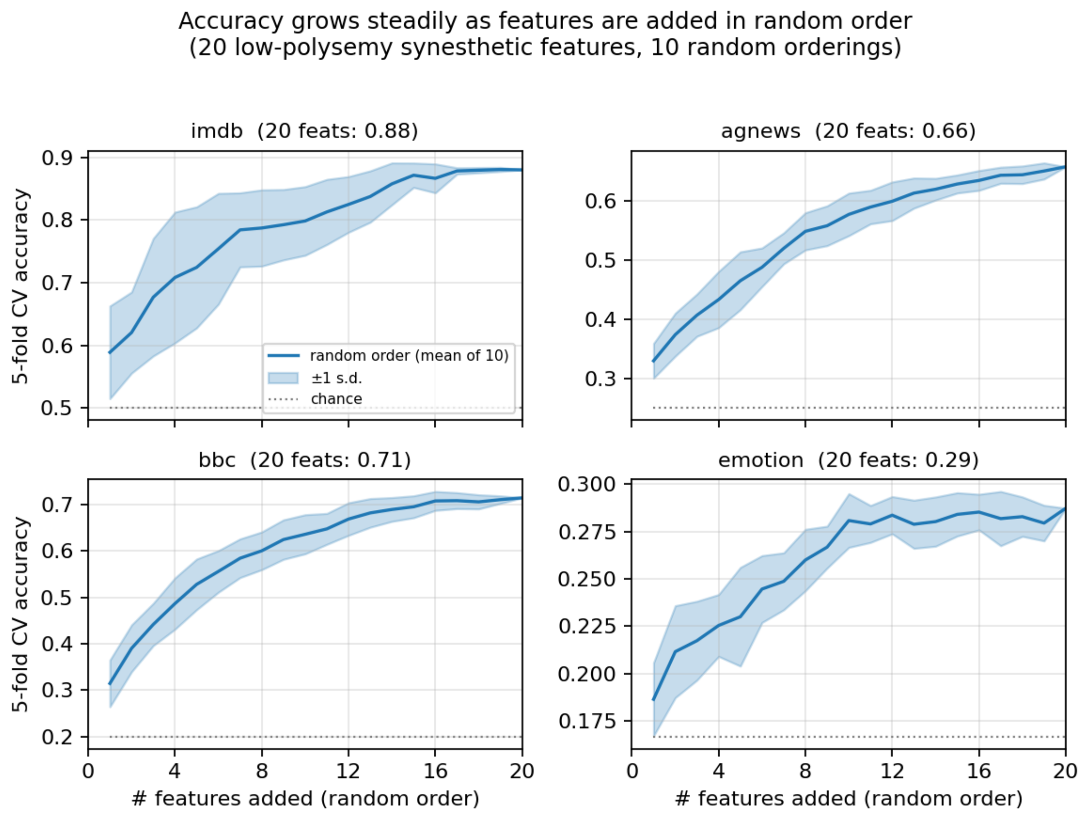

Figure 4: Accuracy as the 20 low-polysemy synesthetic features are added in a *random* order (10 random orderings; mean and ± 1 s.d. band). The steady rise—almost every single-feature addition improves accuracy (100% of steps on AG News, 95% on BBC, 89% on IMDB)—indicates that the features are largely complementary: each one tends to contribute information not already captured by the others.

5 Discussion: there is no referent

Why should "is this text purple?" carry information about whether a review is positive? Our answer is semiotic rather than perceptual.

For de Saussure (1916), a sign takes its value from its differential position in the language system, and the referent—the thing in the world—is set aside. A model trained only on text is the limiting case of this picture: it has access to signifiers and their distributional relations (Firth, 1957), and to nothing else. It has no referent. Consequently, a sensory predicate is not constrained by any perceptual fact. "Purple", "acidic" and "rough" are not, for the model, attempts to report a perception; they are signifiers that acquire a stable value purely through their differences from other signifiers. Asking the model whether a text is purple is therefore not a category error from its point of view: it is a well-formed question about position in *la langue*, and it receives a graded, text-dependent answer.

This reframes what "synesthesia" means here. Synesthesia presupposes *bridges* between modalities; but the model has only one modality (text), so there are no modalities to bridge. What looks like synesthesia is the collapse of every sensory axis onto a single semiotic plane, where "red", "acid" and "rough" are signifiers like any other. The senses place no constraint on the language of the model: a red text is exactly as sayable as an acidic or a smooth one.

Two observations support this reading rather than a "it is just lexicalized metaphor" alternative. First, the effect survives when figurative meaning is removed: low-metaphoricity predicates still classify (Figure 3). Second, the information is not generic—gibberish probes, which are also signifiers but lack a richly structured differential value, are markedly weaker (Figure 1). The value of "purple" comes neither from a referent (there is none) nor necessarily from a dead metaphor, but from the structured place the word occupies in the system. We do not claim the model escapes language: the consistency of its answers may itself reflect humans' own synesthetic usage in the training corpus—there is, in a sense, no outside-the-text (Derrida, 1967). That is precisely the point.

6 Conclusion

Twenty questions about the taste, colour and texture of a text are enough to classify it well above chance and a strong random-word control, with only a weak dependence on whether the sensory words carry figurative meaning. We read this as evidence that, for a referent-free model, sensory predicates are unconstrained signifiers whose usefulness derives from differential structure alone—an empirical illustration of the Saussurean sign without a referent.

Limitations

We study one model family and four English datasets with 500-example subsets; absolute numbers will vary with model and sample size. Our metaphoricity measures (author a-priori and model self-rating) are proxies and could be complemented by human norms such as the Lancaster Sensorimotor Norms (Lynott et al., 2020). The gibberish control equalizes "probe-ness" but not lexical frequency or syntactic form. Finally, we cannot fully separate "no referent" from "the training corpus already contains human synesthetic usage"; we take this entanglement to be constitutive of the phenomenon rather than a confound to be removed.

References

Emily M. Bender, Timnit Gebru, Angelina McMillan-Major, and Shmargaret Shmitchell. 2021. On the dangers of stochastic parrots: Can language models be too big? In *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency (FAccT)*, pages 610–623.

Emily M. Bender and Alexander Koller. 2020. Climbing towards NLU: On meaning, form, and understanding in the age of data. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 5185–5198.

Yonatan Bisk, Ari Holtzman, Jesse Thomason, Jacob Andreas, Yoshua Bengio, Joyce Chai, Mirella Lapata, Angeliki Lazaridou, Jonathan May, Aleksandr Nisnevich, Nicolas Pinto, and Joseph Turian. 2020. Experience grounds language. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 8718–8735.

Richard E. Cytowic. *Synesthesia: A Union of the Senses*. MIT Press.

Ferdinand de Saussure. 1916. *Cours de linguistique générale*. Payot, Paris. Edited by C. Bally and A. Sechehaye.

Jacques Derrida. 1967. *De la grammatologie*. Les Éditions de Minuit, Paris.

John R. Firth. 1957. A synopsis of linguistic theory 1930–1955. In *Studies in Linguistic Analysis*, pages 1–32. Blackwell.

Dorian Florescu and Matthew England. Algorithmically generating new algebraic features of polynomial systems for machine learning.

Gemma Team, Google DeepMind. 2026. Gemma 4 technical report. *arXiv preprint*. Model: `google/gemma-4-E4B-it`, Hugging Face.

Stevan Harnad. 1990. The symbol grounding problem. *Physica D: Nonlinear Phenomena*, 42(1-3):335–346.

Dermot Lynott, Louise Connell, Marc Brysbaert, James Brand, and James Carney. 2020. The lancaster sensorimotor norms: Multidimensional measures of perceptual and action strength for 40,000 english words. *Behavior Research Methods*, 52:1271–1291.

Francesca Strik Lievers. 2015. Synaesthesia: A corpus-based study of cross-modal directionality. *Functions of Language*, 22(1):69–95.

Stephen Ullmann. 1957. *The Principles of Semantics*. Blackwell, Oxford.

Bodo Winter. 2019. *Sensory Linguistics: Language, Perception and Metaphor*. John Benjamins, Amsterdam.

A Question sets

Low-polysemy (per sense). *Sight:* orange, purple, angular, opaque. *Taste:* salty, umami/savory, starchy, minty. *Touch:* wet, grainy, fuzzy, spongy. *Hearing:* muffled, echoey, nasal, squeaky. *Smell:* metallic, smoky, floral, musty.

High-polysemy (per sense). *Sight:* dark, bright, colorful, black. *Taste:* sweet, bitter, sour, bland. *Touch:* warm, cold, rough, smooth. *Hearing:* loud, quiet, shrill, harmonious. *Smell:* stinking, fragrant, fishy, fresh.

Form. Each question is asked as: "*[text]*\n\nQuestion: Is this text *[predicate]*?\nAnswer with only 'yes' or 'no'." and the feature is $\mathrm{logit}(\mathrm{yes}) - \mathrm{logit}(\mathrm{no})$.